

Caesar Tuguinay

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EDUCATION

PhD Electrical Engineering, University of Washington	2025 - 2030 (expected)
Advisor: Dr. Wu-Jung Lee	
MSc Applied and Computational Mathematics, University of Washington	2024 - 2025
BS Mathematics, University of Washington	2019 - 2023

RESEARCH EXPERIENCE

Graduate Research Assistant, Applied Physics Lab, University of Washington	2024 - present
Research Assistant, Applied Physics Lab, University of Washington	2023 - 2024
Undergraduate Research Assistant, Speech and Hearing Sciences Department, University of Washington	2023

RESEARCH PUBLICATIONS

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- Tuguinay C.**, Lee W.J., Staneva V., Phillips E.M., Clemons J, Billings A. (2026). "An open-source, scalable workflow for organizing echosounder data for machine learning applications." [Submitted for publication].
- Lee W.J., **Tuguinay C.**, Staneva V., Izard L., Butala S., Lucca B.M., Billings A., Phillips E.M., Pohl J.E., Clemons J. (2026). "An edge-to-cloud pipeline for near-real-time biological interpretation of echosounder data." [Manuscript in preparation].
- Lee W.J., **Tuguinay C.**, Staneva V. (2026). "Semantic segmentation of multi-frequency echograms for Pacific hake aggregations with diffused boundaries." [Manuscript in preparation].
- Lee W.J., Setiawan L., **Tuguinay C.**, Mayorga E., Staneva, V. (2024). "Interoperable and scalable echosounder data processing with Echopype." *ICES Journal of Marine Science*, 0(0).
- Lee W.J., Staneva V., Setiawan L., Mayorga E., **Tuguinay C.**, Butala, S., Lucca B., Lei D. (2024). "Echostack: A flexible and scalable open-source software toolbox for echosounder data processing." *2024 SciPy Proceedings*.

CONFERENCE PRESENTATIONS

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- Tuguinay C.**, Lee W.J. "Investigating internal representations of echogram semantic segmentation models using non-negative matrix factorization." *OCEANS 2026 workshop at Simon Fraser University*, July 2026. Burnaby, Canada.
- Tuguinay C.**, Lee W.J. "Exploring context relevance for echogram classification tasks using class activation map methods." *The Acoustical Society of America 2026 Spring Meeting*, May 2026. Philadelphia, Pennsylvania.
- Tuguinay C.**, Lee W.J., Staneva V. "An open-source workflow for organizing fisheries acoustics data from transect surveys for machine learning applications." *ICES Working Group on Fisheries Acoustics, Science and Technology 2026*, April 2026. Wageningen, Netherlands.
- Lee W.J., Staneva V., **Tuguinay C.**, Butala S., Lucca B., "High-throughput information processing in fisheries and plankton acoustics." *The Acoustical Society of America 2025 Spring Meeting*, May 2025. New Orleans, Louisiana.
- Tuguinay C.**, Lee W.J., Staneva V. "An open-source workflow for organizing fisheries acoustics data from transect surveys for machine learning applications." *The Acoustical Society of America 2025 Spring Meeting*, May 2025. New Orleans, Louisiana.

Lee W.J., Staneva V., Setiawan D., **Tuguinay C.**, Butala S., Lucca B., Dingrui L. “Echostack: A flexible and scalable open source software suite for echosounder data processing.” *SciPy 2024*, July 2024. Tacoma, Washington.

Lee W.J., Staneva V., **Tuguinay C.** “Variability and influence of fisheries acoustic echogram annotations on machine learning applications.” *The Acoustical Society of America 2024 Spring Meeting*, May 2024. Ottawa, Canada.

Cler G.J., **Tuguinay C.**, Sokolov S., Bullen A. “Acoustic factors predicting listener perception of stuttering.” *Conference on Motor Speech*, February 2024. San Diego, California.

OPEN-SOURCE SOFTWARE CONTRIBUTIONS

Echopype (*developer*): Python package designed to parse, calibrate, regrid, and perform other scientific computations on large amounts of raw acoustic data from various commercial and scientific Echosounder brands.

Echoregions (*developer and maintainer*): Python package that interfaces with annotations of water column sonar data for training machine learning models or doing other downstream processing such as biomass estimation.

HONORS AND AWARDS

University of Washington Undergraduate Dean’s List	2019-2023
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ACADEMIC SERVICE

TA and lecturer for the OOIFB Summer School on Acoustics	July 14-25 2025
TA for the New Orleans Bridge to Ocean Acoustics Program workshop	May 17-18 2025
TA for the Seattle Bridge to Ocean Acoustics Program workshop	March 24-25 2025

FIELD EXPERIENCE

2026 Integrated West Coast Pelagic Survey Research Leg 5 (*17 days offshore*): Collected broadband and narrowband single-target acoustic backscatter data of mesopelagic and semi-demersal fish using vertically and horizontally oriented echosounders mounted on a CTD-rosette.

2025 Integrated West Coast Pelagic Survey Research Leg 6 (*9 days offshore*): Collected broadband and narrowband acoustic backscatter data from fish schools using a broadband echosounder deployed as a water column profiler.

AFFILIATIONS

Acoustical Society of America, student member	2026 - present
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